

MKUFT Core Extended

Professional reading edition of the evolving public MKUFT canon

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Purpose

The McLaughlin–Kairos Unified Field Theory (MKUFT) models reality through a typed architecture of substrate possibility, information structure, physical dynamics, and observer-positioned registration.

The four-layer shorthand is

$$S \rightarrow I \rightarrow P \rightarrow O.$$

This core document states the architecture. It does not claim that the physical dynamics required for a complete unified theory have already been derived.

Physics positioning

The term **Unified Field Theory** carries an immediate scientific burden. A viable foundational programme must address the relationship between quantum theory and gravitation, including entanglement, Bell-compatible correlations, no-signalling, quantum-field-theory limits, and recovery of general relativity.

MKUFT has not yet completed those derivations.

The proposed foundational reframing is developed in [Layer Before Law](#). That module asks whether quantum and gravitational behaviour may be effective P-layer outputs of different relational conditions within a deeper lawful S–I–P–O architecture.

The proposal fails as a foundational physical theory if it cannot recover established physical limits, preserve Bell compatibility and no-signalling, or distinguish its mechanism from existing formalisms and interpretations.

This core should therefore be read as the architecture that any successful physics-facing derivation would have to make lawful, not as a substitute for that derivation.

Layer architecture

S — Substrate

The substrate is the proposed possibility or source domain in which candidate configurations are represented. It is not directly observed and should not be treated as a detected material medium merely because it has a mathematical representation.

I — Information

The information layer carries structured correlations, constraints, patterns, address, state, and proto-physical forms. Informational structure does not become physical mechanism without a defined coupling.

P — Physical

The physical layer contains realised events, measurements, trajectories, matter, energy, fields, instruments, and other measurable outcomes. Standard physics remains the baseline in its demonstrated domains.

O — Observer

The observer layer contains observer-positioned registration, measurement context, record, attention, interpretation, memory, salience, and bounded state-dependent participation where operationally defined.

The observer is not an unlimited creator of outcomes and cannot replace missing physical dynamics.

Core claim

MKUFT does not reject standard physics. It requires standard physics to emerge as the dominant limiting description where additional substrate, information, or observer terms add no measurable predictive value.

A stronger MKUFT claim is possible only if a deeper architecture eventually recovers established quantum and gravitational regimes from one lawful mechanism rather than merely placing them side by side.

Any claimed deviation from accepted physical prediction must be small, controlled, preregistered, and independently testable. The framework is not a licence for unconstrained anomaly.

Observer coherence

An observer state may be represented by ρ_O , with a bounded coherence functional

$$\kappa(\rho_O) \in [0, 1].$$

High κ denotes a more internally aligned or lower-noise state according to a declared proxy; low κ denotes a more fragmented or noisy state.

κ requires operational definition in each experiment. Observer coherence does not create arbitrary outcomes. Any claimed physical modulation requires an independently specified interaction term and must survive apparatus, environmental, expectancy, and statistical controls.

Composite-observer candidates

The O-layer may be addressed at more than one scale, but a collection of observers does not automatically become a larger observer.

A reciprocally coupled system may be treated as a **composite-observer candidate** only where the coupled level demonstrates a sufficiently persistent boundary, retained joint state, reciprocal causal influence, system-level discrimination, correction propagation, and closed action–measurement–feedback loops.

The stronger superadditivity claim requires a matched comparison. A provisional gain is

$$G_{\text{syn}} = F_{\text{coupled}} - F_{\text{add}},$$

where F_{add} is the strongest matched additive or independent-ensemble baseline appropriate to the task.

For example, if

$$F_A = 2, \quad F_B = 2, \quad F_{\text{add}} = 4, \quad F_{\text{coupled}} = 6,$$

then

$$G_{\text{syn}} = 2.$$

The evidential object is the positive excess, not merely the fact that the output belongs to the coupled system.

This remains a bounded functional hypothesis. It does not establish subjective consciousness, merged identity, personhood, a new physical force, direct substrate coupling, or a fifth fundamental MKUFT layer.

The operational criteria and stronger controls are developed in [Cross-Layer Invariants and Layer Addressing](#), [Active Traversal and Functional Emergence](#), and [Strongest Fair Null and Relational Specificity](#).

Realisation logic

For event E , the realised probability scaffold combines standard physical dynamics with candidate substrate-to-information and observer-condition terms. The fuller mathematical treatment is given in the [Mathematical Appendix](#).

A compact limiting statement is

$$P_{\text{realized}}(E) \approx P_{\text{phys}}(E)$$

whenever additional terms become constant, negligible, or otherwise reduce to the established physical distribution within a declared regime and tolerance.

[Layer Before Law](#) introduces a separate deeper task: define a lawful complete-state update capable of recovering both quantum and gravitational effective limits. The notation $\text{Update}_{\text{SIP0}}$ names that missing task; it is not already a derived law.

Standard-physics limit

The ordinary limit is mandatory. When observer-linked or substrate-information terms add no predictive structure, MKUFT must reduce to the relevant accepted physical model.

For the foundational branch, this includes recovery of established quantum and gravitational predictions in the regimes where those theories are experimentally supported.

A failure of ordinary-limit recovery is a failure of the corresponding physics-facing branch.

Falsifiability

Candidate tests include:

- REG/RNG comparisons under preregistered observer-state conditions;
- strictly blinded remote-information protocols;
- preregistered co-occurrence or synchronicity-density studies;
- environmental modulation studies with measured noise, shielding, geometry, or material conditions;
- group-coherence tests under matched controls;
- composite-observer superadditivity tests against independent, one-way, state-reset, replay, and relation-scrambled controls;
- controlled geometry and boundary-response studies;
- Bell-compatible and no-signalling derivation tests for any proposed entanglement mechanism;
- quantum and gravitational recovery tests for any proposed complete S-I-P-O update rule.

These are research programmes, not established confirmations.

Repeated failure under strong controls weakens or rejects the relevant branch. Failure to recover established quantum or gravitational limits falsifies the corresponding foundational physics claim even if other MKUFT methods remain useful in non-foundational domains.

Related public documents

- [Layer Before Law](#)
- [Mathematical Appendix](#)
- [Standalone Formal Addendum](#)
- [Experimental Test Programme](#)
- [Falsification Summary](#)
- [Typed Traversal and Equation Hygiene](#)
- [Scientific Tightening and Claim Discipline](#)